

Optimising prediction of early metastasis-free survival in uveal melanoma using a four-category model incorporating gene expression profile and tumour size

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ABSTRACT

Background Largest basal diameter (LBD) appears to have independent prognostic value in uveal melanoma (UM).

Methods All patients undergoing plaque brachytherapy or enucleation for UM involving the choroid and/or ciliary body between 2012 and 2019.

Results A total of 348 patients with a mean age of 60±14 years were included and followed for a mean of 40±26 months (3.3±2.2 years). On multivariate analysis, LBD >12 mm remained a significant independent predictor of metastasis for both class 1 (HR 21.90; 95% CI 2.69 to 178.02; p=0.004) and class 2 (HR 2.45; 95% CI, 1.03 to 5.83; p=0.04) tumours. Four prognostic groups were created: group 1 (class 1, LBD <12 mm), group 2 (class 1, LBD ≥12 mm), group 3 (class 2, LBD <12 mm) and group 4 (class 2, LBD ≥12 mm). Life tables were used to calculate the 3-year and 5-year metastasis-free survival: group 1 (98 and 98%), group 2 (86 and 86%), group 3 (81 and 62%) and group 4 (54 and 47%). Compared with the reference category (group 1), the Cox proportional hazard model demonstrated a significant worsening of survival for each progressive category (group 2 (HR 21.59; p=0.004), group 3 (HR 47.12, p<0.001), and group 4 (HR 114.24; p<0.001)). In our dataset, the four-category Cox model performed poorer compared with the American Joint Committee on Cancer (AJCC) and gene expression profile (AJCC+GEP) in the Akaike's information criteria (AIC) (297 vs 291), fit better with the Bayesian information criteria (BIC) (309 vs 313) and performed similarly with the Harrel's C (0.86 (95% CI 0.80 to 0.91) vs 0.89 (0.84 to 0.94), respectively).

Conclusions Combination of GEP and LBD allows separation of patients into four easy-to-use prognostic groups and was similar to a model combining AJCC stage with GEP.

involvement and extraocular extension, form the basis of the American Joint Committee on Cancer (AJCC) staging system,¹⁰ which has been validated in two large cohorts.^{11 12}

Over the past few decades, several advances in the molecular genetics of UM have refined not only our understanding of the disease pathophysiology, but also our prognostication capabilities. A variety of platforms for chromosomal analysis have been described.^{13–19} In addition to examining tumour DNA, the genetic microenvironment of a lesion can be quantified via RNA analysis using a 15-gene assay to provide a gene expression profile (GEP) classification.²⁰ This test categorises patients as either low (class 1) or high risk (class 2) for developing metastatic disease within 3–5 years²⁰ and has been prospectively validated in a multicentre study.²¹ As GEP has been investigated with longer follow-up, identification of a late metastasis group allowed for further stratification of class 1 tumours into class 1A (lowest metastatic risk) and class 1B (moderate long-term metastatic risk) based on the level of CDH1/RAB31 expression.²²

Initial studies evaluating GEP, primarily on large tumours treated with enucleation, found that although a class 2 result was strongly correlated with various clinical and cytogenetic factors portending a poor prognosis, none of these variables contributed additional prognostic information independent of GEP.²¹ As subsequent studies included tumours undergoing plaque brachytherapy in addition to enucleation, LBD emerged as a second independent prognostic factor.^{23 24} While class 2 UMs have a significant improvement in prognosis when their LBD is <12 mm at the time of treatment, it is unclear whether this finding remains significant for class 1 tumours.²⁴ Improvements in biopsy technique have led to increased comfort sampling smaller tumours that historically would not have had genetic information available.

The concept that combining genetic data with clinical features could improve prognosis is not a new one. In 2007, Damato *et al* reported that prediction of disease-specific mortality was improved when cytogenetic analysis was interpreted alongside tumour diameter and cell type.¹⁶ Although the current eighth edition AJCC staging system does not incorporate results of genetic testing, several groups have reported an improved prognostic value when AJCC stage is combined with chromosome 3 and 8q status.^{25–27} Additionally, prognostic tools

Although the past four decades have seen a significant shift towards eye-conserving treatments for uveal melanoma (UM), the 5-year survival rate has remained largely unchanged.¹ The optimal method by which to estimate the risk of metastasis in patients with UM remains an ever evolving and ongoing debate in the field of ocular oncology. Several patient features,^{2 3} tumour characteristics^{4 5} and histopathological findings^{6–9} have previously been identified as predictors of metastasis and/or survival in UM. Anatomical factors, including tumour thickness, largest basal dimension (LBD), ciliary body



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including the Liverpool Uveal Melanoma Prognosticator Online (LUMPO)^{28 29} and PRiMeUM³⁰ aim to provide personalised risk estimates by taking both clinical and chromosomal information into consideration. More recently, The Cancer Genome Atlas (TCGA) classification was developed^{31 32} and the authors concluded that it provides accuracy that is superior to AJCC for predicting metastasis from UM.³³

The purposes of this study were (1) to validate the findings of previous studies^{23 24} on a similarly large, but independent cohort of patients, (2) to clarify the prognostic importance of LBD in class 1 tumours and (3) to describe and compare the performance of GEP class and LBD versus AJCC eighth edition versus AJCC+GEP, as this has not previously been reported. As the current study included many small tumours, it likely better reflects the full spectrum of patients undergoing treatment for UM.

METHODS

Following ethics approval from the Health Research Ethics Board for Alberta – Cancer Committee, data were collected from a prospective database of patients undergoing treatment for UM and testing for GEP at our centre between July 2012 and December 2019. Patients with UM involving the choroid and/or ciliary body were included. Subjects were excluded if their biopsy did not yield a successful GEP result. Baseline demographic data, tumour-specific variables and AJCC eighth edition classification were recorded. All patients included in this study were screened according to our standardised consensus-based provincial clinical practice guidelines (low-risk patients received an abdominal ultrasound annually, high-risk patients received imaging every 6 months consisting of a liver ultrasound alternating with abdominal MRI).³⁴ Low risk was defined as class 1A or 1B and <8 mm in thickness, and high risk was class 2 or ≥8 mm in thickness. Metastatic and survival data were obtained by reviewing a province-wide electronic repository containing the results of all surveillance diagnostic imaging. For out-of-province patients, the patients and/or their referring providers were contacted to confirm their metastatic/survival status.

Fine needle aspiration biopsy (FNAB) for GEP was performed at the time of plaque insertion using a 27-gauge long needle attached to intravenous extension tubing on a 10 mL syringe. A trans-scleral approach was used for anteriorly located tumours, while for posteriorly located lesions, a trans pars-plana trans-vitreous approach under visualisation with indirect ophthalmoscopy and a 20D lens was performed. All tumours treated with brachytherapy were prescribed a treatment dose of 70 Gy to the apex which was calculated with modelling to account for attenuation through the silastic and reflection off the back wall of the plaque. Enucleation specimens were formalin fixed and an unstained tissue slide was then sent for GEP assessment. A GEP class designation was provided by Castle Biosciences using a commercially available assay as has been previously described.²¹

Descriptive statistics were used, followed by univariate and multivariate Cox proportional hazard analyses to assess for predictors of metastasis and death. Statistical significance was defined as $p < 0.05$. To keep the analysis broad, variables in the univariate analysis with a $p < 0.1$ were employed in the multivariable analysis, and criteria for removal from the model were based on the insignificance of the variable's p value. Stepwise regression using backward elimination was used to develop the final multivariable model by removing insignificant variables one at a time. Proportional hazards were evaluated visually evaluating the Schoenfeld residuals and the Schoenfeld residual statistical

test. Kaplan-Meier curves were constructed to provide a visual estimate of survival and metastatic risk based on key significant variables. Based on prior literature^{23 24} an a priori plan, and confirmation from the initial findings of our analysis, patients were stratified into four groups based on a combination of GEP class and a measure of tumour size (LBD). Internal validation was then performed using a bootstrap analysis to assess statistical precision and to evaluate the level of overfitting. A 100-step bootstrap with a change in the HR or statistical significance for the variable was defined as overfitting.

In our opinion, the proposed four group model is simplistic enough to be used at the point of care and to be understood by patients (personal experience, EW). This four-group classification scheme was assessed as a Cox regression model using a four-level categorical variable with metastasis as the outcome. To better understand the trade-off of using this model, it was compared with more complicated models including the AJCC eighth edition staging system, the AJCC eighth edition staging system (I, IIA, IIB, IIIA, IIIB, IIIC) combined with GEP class (1 or 2), a simplified three-stage AJCC model (I, II and III) and a multivariate model (that would require an online calculator) built to maximise predictability by including all significant predictors used as continuous variables. Due to the varying emphasis and penalisation methods, multiple accepted analysis methods for evaluating the performance of the models including the Akaike's information criteria (AIC), Bayesian information criteria (BIC) and Harrell's C were used. This research adhered to the tenets of the Declaration of Helsinki.

RESULTS

A total of 348 patients with posterior UM with a mean age of 60 ± 14 years (range 20–93 years) and a mean follow-up of 40 ± 26 months (3.3 ± 2.2 years) were included. Roughly half were female (182/348; 52%). The mean tumour thickness was $4.9 \text{ mm} \pm 3.0 \text{ mm}$ (range = 0.0–15.8 mm) and mean LBD was $10.0 \text{ mm} \pm 4.2 \text{ mm}$ (range = 2.5–22.0 mm). Only 15% (53/348) of patients had ciliary body involvement, and extraocular extension was seen in 1% (5/348). Nearly half of the patients were AJCC T1a (164/348; 47%) one-quarter were T2a (85/348; 24%) and 59 cases were AJCC T3 (59/348; 17%) (table 1). Over the study period, a total of 41 patients developed metastatic disease from UM (41/348; 12%) and 29 of these patients died as a result of their metastatic disease (29/348; 8%). There were eight additional patients who died from other causes during the study period (8/348; 2%) without detection of metastatic disease on surveillance imaging prior to their death.

FNAB yielded a successful GEP classification in 98% of patients undergoing plaque brachytherapy (321/326) and 100% of those undergoing enucleation (22/22) with multiple gene failure occurring in only 1% of all cases overall (5/348). The five patients without a GEP result were excluded from the survival analysis. Nearly half of the tumours in the study were class 1A (160/343; 46%) followed by a similar proportion of class 1B (95/343; 28%) and class 2 (88/343; 26%) lesions. There were no significant differences in demographic or baseline characteristics between class 1A and 1B; however, class 2 lesions were different in several ways. Patients with class 1B tumours were slightly younger (mean age 62.1 ± 14.9 years) than those with class 2 (mean age 63.6 ± 15.9 years) ($p < 0.001$). Class 2 tumours were thicker ($7.0 \pm 3.5 \text{ mm}$) than either class 1A ($4.1 \pm 2.2 \text{ mm}$; $p < 0.001$) or class 1B ($4.2 \pm 2.8 \text{ mm}$; $p < 0.001$) and had a greater mean LBD (12.6 ± 4.1 vs class 1A (9.2 ± 3.6) vs class 1B (9.1 ± 4.4); $p < 0.001$). As would be expected, the proportion of

Table 1 Baseline demographic and clinical features, AJCC stage of 348 patients with uveal melanoma subdivided by GEP class

	All patients (n=348)*	Class 1A (n=160)	Class 1B (n=95)	Class 2 (n=88)	1A vs 1B p value	1A vs 2 p value	1B vs 2 p value
Gender (%)					0.31	0.68	0.6
Male	167 (48)	72 (45)	49 (52)	42 (48)			
Female	182 (52)	88 (55)	46 (48)	46 (52)			
Age† (years)	60±14 (mean±SD, range)	60±15 (20–93)	56±16 (21–90)	63±12 (32–87)	0.29	0.84	<0.001
Thickness† (mm)	4.9±3.0 (mean±SD, range)	4.1±2.2 (0.0–15.8)	4.2±2.8 (1.1–12.6)	7.0±3.5 (2.0–15.8)	0.71	<0.001	<0.001
Largest basal dimension†	10.0±4.2 (mean±SD, range)	9.2±3.6 (3.0–20.0)	9.1±4.4 (2.5–22.0)	12.6±4.1 (2.7–20.0)	0.79	<0.001	<0.001
Ciliary body involvement† (%)	53 (15)	19 (12)	8 (8)	26 (30)	0.38	0.001	0.001
Extraocular extension‡	5 (1)	2 (1)	0 (0)	3 (3)	–	0.27	–
AJCC T category† (%)					0.81	<0.001	<0.001
T1a	164 (47)	90 (56)	55 (58)	17 (19)			
T1b	9 (3)	4 (2)	1 (1)	4 (5)			
T1c	0 (0)	0 (0)	0 (0)	0 (0)			
T1d	0 (0)	0 (0)	0 (0)	0 (0)			
T2a	85 (24)	38 (24)	20 (21)	24 (27)			
T2b	11 (3)	7 (4)	2 (2)	2 (2)			
T2c	0 (0)	0 (0)	0 (0)	0 (0)			
T2d	0 (0)	0 (0)	0 (0)	0 (0)			
T3a	41 (12)	11 (7)	11 (12)	19 (22)			
T3b	15 (4)	5 (3)	4 (4)	6 (7)			
T3c	0 (0)	0 (0)	0 (0)	0 (0)			
T3d	3 (1)	1 (1)	0 (0)	2 (2)			
T4a	5 (1)	2 (1)	1 (1)	2 (2)			
T4b	13 (4)	1 (1)	1 (1)	11 (13)			
T4c	0 (0)	0 (0)	0 (0)	0 (0)			
T4d	2 (1)	1 (1)	0 (0)	1 (1)			
T4e	0 (0)	0 (0)	0 (0)	0 (0)			
AJCC stage†					0.99	<0.001	<0.001
Stage I	164 (47)	90 (56)	55 (58)	17 (19)			
Stage IIA	94 (27)	42 (26)	21 (22)	28 (32)			
Stage IIB	50 (14)	17 (11)	12 (13)	21 (24)			
Stage IIIA	22 (6)	8 (5)	6 (6)	8 (9)			
Stage IIIB	16 (5)	2 (1)	1 (1)	13 (15)			
Stage IIIC	2 (1)	1 (1)	0 (0)	1 (1)			
Metastasis	41 (12)	9 (6)	2 (2)	30 (34)	0.2	<0.001	<0.001
Follow-up (months)	40±26 (mean±SD, range)	41±27 (0–93)§	42±25 (4–93)§	28±23 (0–93)§	0.72	<0.001	<0.001

*Patients could not be classified by GEP due to multiple gene failure on the FNAB specimen submitted for analysis.

†Significant result.

‡No cases of extraocular extension in this series were >5 mm.

§Two patients had <1 month follow-up: one class 1A patient died from metastatic lung cancer <1 month following plaque brachytherapy; one class 2 patients died from widespread UM metastatic disease within 1 month of plaque brachytherapy. Both of these patients were AJCC T2a. All other patients had a minimum of 3 months of follow-up. AJCC, American Joint Committee on Cancer; FNAB, fine needle aspiration biopsy.

higher AJCC T category and stage was greater in patients with class 2 versus either class 1A or 1B ($p<0.001$) (table 1). PRAME results were only available for 36% (58/160) of class 1A, 41% (39/95) of class 1B and 41% (36/88) of class 2 tumours as this test was not commercially available in the early enrolment period.

Univariate Cox proportional hazard analysis for metastasis found several significant predictors including tumour thickness ($p<0.001$), LBD ($p<0.001$), ciliary body involvement ($p=0.001$) and extraocular extension ($p<0.001$). The potential for a non-linear relationship was evaluated by inserting the squared term of thickness and largest basal dimension in the models. These were both found to be insignificant, suggesting

a linear relationship. Compared with their respective reference groups (GEP class 1A and AJCC stage I), GEP class 2 ($p<0.001$) and AJCC IIB ($p<0.001$) IIIA ($p=0.002$), IIIB ($p<0.001$) and IIIC ($p<0.001$) were also significant predictors (HR and 95% CI reported in table 2). Although AJCC Stage IIA did not achieve our statistical definition of significance, the effect size imparted by a HR of 3.79 is clinically significant and it is possible that our study was underpowered to find a difference compared with the reference category of stage 1. Graphical evaluation of the Schoenfeld residuals and statistical testing demonstrated no violation of the proportional hazard assumption. Metastasis-free survival was also assessed

Table 2 Significant variables on univariate Cox proportional hazard analysis for predictors of metastasis.

	Metastasis	
	HR (95% CI)	P value
Enucleation	4.46 (1.94–10.27)	<0.001
Ciliary body involvement	4.18 (2.17–8.51)	<0.001
Extraocular extension	17.18 (4.93–59.94)	<0.001
Tumour thickness*	1.36 (1.24–1.50)	<0.001
Thickness 3.0–8.0 mm	4.15 (1.27–24.09)	0.023
Thickness >8.0 mm	23.68 (5.43–103.21)	<0.001
Largest basal diameter*	1.29 (1.20–1.40)	<0.001
LBD >12.0 mm	7.77 (3.51–17.16)	<0.001
GEP		
Class 1A (reference)		
Class 1B	0.54 (0.11–2.67)	0.449
Class 2	11.28 (4.62–27.53)	<0.001
GEP combined with LBD		
Group 1 (GEP class 1 and LBD <12 mm) (reference)		
Group 2 (GEP class 1 and LBD >12 mm)	21.59 (2.66–175.51)	0.004
Group 3 (GEP class 2 and LBD <12 mm)	47.12 (5.79–383.70)	<0.001
Group 4 (GEP class 2 and LBD >12 mm)	114.24 (15.26–855.09)	<0.001
AJCC stage		
Stage I (reference)		
Stage IIA	3.79 (0.98–14.66)	0.054
Stage IIB	14.64 (4.06–52.85)	<0.001
Stage IIIA	10.16 (2.27–45.42)	0.002
Stage IIIB	44.30 (11.18–175.51)	<0.001
Stage IIIC	282.84 (43.36–1845.19)	<0.001

*Both tumour thickness and largest basal diameter were analysed both as continuous and categorical variables.

†Significant result.

using Kaplan-Meier survival estimates for GEP class and LBD <12 mm or ≥12 mm (figure 1).

Due to the lack of significant differences in class 1A and 1B tumours in our series, class 1A and 1B were combined for the multivariable analysis. On multivariable Cox proportional hazard analysis, GEP class 2 (HR 9.09; 95% CI 3.99 to 20.72; $p < 0.001$) and LBD >12 mm (HR 4.16; 95% CI 1.84 to 9.44; $p = 0.001$) remained important predictors of metastasis-free survival. In addition, LBD >12 mm remained a significant

independent predictor of metastasis for both class 1 (HR 21.90; 95% CI, 2.69 to 178.02; $p = 0.004$) and class 2 (HR 2.45; 95% CI, 1.03 to 5.83; $p = 0.04$) tumours.

After including GEP, thickness and LBD in the multivariable model, the association between tumour thickness and metastasis became insignificant. As such, tumour thickness was removed from the final model. A 100-repetition bootstrap was performed on this multivariate Cox proportional hazards model to assess statistical precision and overfitting. All HRs did not change and statistical results remained significant. The combination of GEP class with LBD results in four distinct prognostic groups as follows: group 1 (GEP class 1 and LBD <12 mm), group 2 (GEP class 1 and LBD ≥12 mm), group 3 (GEP class 2 and LBD <12 mm) and group 4 (GEP class 2 and LBD ≥12 mm). The multivariate model with four groups was assessed for goodness of fit with the likelihood ratio test, which demonstrated the model fit the data well (LR $\chi^2 = 65.32$, $p < 0.0001$).

Life tables were used to calculate the 3-year and 5-year metastasis-free survival (95% CI) which was (98% (95% to 99%) and 98% (95%–99%)) for group 1, (86% (73% to 93%) and (86% (73% to 93%)) for group 2, (81% (57% to 93%) and 62% (35% to 80%)) for group 3 and (54% (37% to 68%) and 47% (28% to 64%)) for group 4. Compared with the reference category (group 1), the Cox proportional hazard model for the four-category model demonstrated a significant worsening of survival for each progressive category (group 2 (HR 21.59; $p = 0.004$), group 3 (HR=47.12, $p < 0.001$), group 4 (HR=114.24; $p < 0.001$)).

Comparison of the four-category prognostication model to AJCC eighth edition ± GEP class and the maximal regression model revealed that all models behaved similarly. In our dataset, the four-category Cox model predicted UM metastasis significantly better than the AJCC eighth edition in all criteria. The four-category Cox model performed poorer compared with the AJCC + GEP in the AIC, fit significantly better with the BIC and performed similarly with the Harrel's C. The 12-category model produced by combining GEP class (two groups) with AJCC stage (six groups) performed better than AJCC stage alone with both AIC, BIC and Harrel's C. The simplified three-stage AJCC GEP 1 or 2 demonstrated poorer scores on AIC and BIC, but similar scores on the Harrel's C statistic. The complicated regression model, requiring an online calculator, was slightly more predictive than all other models (table 3).

DISCUSSION

Prior studies have found an improvement in prognostication when AJCC and chromosomal analysis are combined.^{25–27}

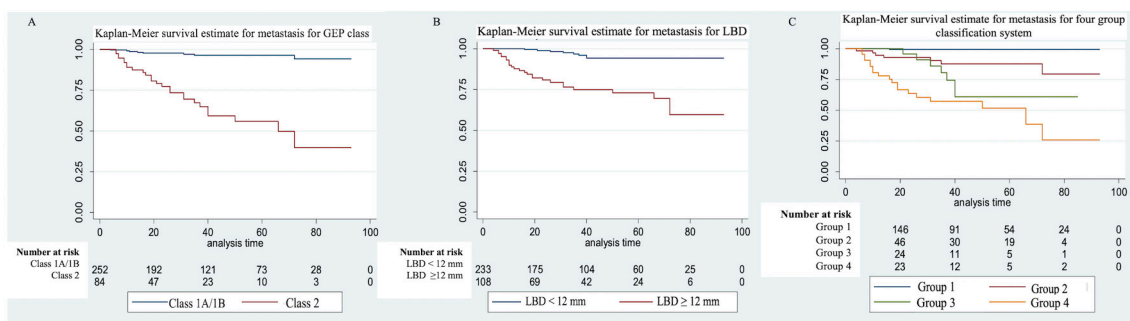


Figure 1 Kaplan-Meier survival estimates demonstrate (A) a poorer prognosis for gene expression profile (GEP) class 2 compared with class 1. Similarly, when all GEP classes are combined, (B) patients with largest basal diameter (LBD) ≥12 mm have a poorer prognosis. Finally, (C) stratification of metastasis-free survival can be predicted using a simple four-group classification.

Table 3 Comparison of the four-group prognostication model to AJCC stage, GEP and AJCC Stage, GEP, LBD, thickness and extraocular extension using Akaike's information criteria, Bayesian information criteria and Harrell's C

Model	Akaike's information criteria (AIC)*	Bayesian information criteria (BIC)*	Harrell's C and 95% CI*
AJCC TNM stage (six levels††)	319	339	0.83 (0.77 to 0.90)
AJCC TNM stage (six levels††)+GEP 1 or 2	291	313	0.89 (0.84 to 0.94)
Simplified AJCC TNM stage (I, II, III)	332	340	0.78 (0.71 to 0.84)
Simplified AJCC TNM stage (I, II, III)+GEP 1 or 2	302	313	0.86 (0.81 to 0.92)
Four-group model	297	309	0.86 (0.80 to 0.91)
GEP, LBD, Thickness and extraocular extension ‡‡	279	294	0.89 (0.84 to 0.95)

*A lower score on the AIC and BIC is a better score. A higher coefficient on the Harrell's C is a better score.

†AJCC stages included were: I, IIA, IIB, IIIA, IIIB, IIIC.

‡The regression model included GEP (class 1 vs class 2), LBD and thickness (both as continuous variables) and extraocular extension (presence vs absence). AJCC, American Joint Committee on Cancer; GEP, gene expression profile; LBD, largest basal diameter; TNM, tumour, node, metastases.

Similarly, addition of tumour size to GEP appears to improve prognostication.^{23 24} Recently, a four-group classification system based on chromosomal analysis was proposed and found to be prognostically superior to AJCC.³³ The purpose of this study was to provide a similar, easy-to-use classification system for clinicians who use GEP rather than chromosomal analysis. Many of the findings of this study, including the high technical success rate (99% overall) and accuracy of GEP as a prognostic test,²¹ importance of tumour size (LBD) as an independent predictor of metastasis,^{23 24} and correlation between higher AJCC Stage and GEP class,³⁵ are in keeping with prior reports. Moreover, as nearly half (45%; 157/348) of the tumours were ≤ 3.5 mm in thickness, the performance of GEP in this cohort of patients supports its use as a reliable and accurate prognostic test for small tumours. Although only 12% of patients developed metastatic disease during the study period, this is in keeping with what would be expected given the high proportion of small, class 1 tumours.

Compared with previously published series where mean tumour thickness and LBD were larger ((5.1 mm and 11.9 mm),²³ (6.6 and 14.6 mm)²⁴ and (6.3 mm and 12.8 mm),²¹ respectively), the tumours included in this study were smaller (4.9 mm and 10.0 mm) (table 4). Prior research has demonstrated that larger tumours are more likely to harbour a class 2 GEP^{23 35 36} or monosomy^{37–39}; however, early reports did not find tumour size a significant predictor of metastasis in class 1 tumours (Walter *et al*).²⁴ With the inclusion of a large proportion of small tumours in this study, we found LBD >12 mm to be a significant independent predictor of metastasis for both class 1 and 2 tumours ($p=0.004$ and 0.043 , respectively). Although controlling for lead-time bias remains a challenge, this suggests that even in less genetically aggressive tumours, treatment at a smaller size results in improved metastasis-free survival. The results of our multivariable Cox proportional hazard models

support this statement by showing that when GEP class is controlled for, at any time during follow-up (a) for every 1 mm increase in thickness, the risk of metastasis increases by 23% or (b) for every 1 mm increase in LBD, the risk of metastasis increases by 21%.

In this study, we were able to use the previously described 12 mm cut-off in LBD to categorise patients into four groups that can easily be applied in real-time clinical practice. In our experience, the simplicity of this categorisation and the significant separation in metastasis-free survival between these four groups make the results of this study quick and easy to incorporate in daily clinical practice and allows for good patient understanding.

Moreover, in our dataset, this four-group model was significantly more predictive than AJCC eighth edition stage and T category. Comparison of the AIC, BIC and Harrell's C of this four-category model to the AJCC eighth edition stage (six groups or simplified 3 groups) combined with GEP class 1 or 2 demonstrated no improvement in performance with the more complicated 6 or 12 category system. Further research will be required to determine the optimal method of prognostication as the AJCC size categories may also evolve. Additionally, we found an improvement in prognostication over AJCC alone when GEP class was combined with AJCC stage, in keeping with previously published reports supporting the addition of genetic data to future editions of AJCC. As we do not routinely perform chromosomal analysis at our institution, we were unable to calculate TCGA for comparison to our four-group classification system. Finally, when interpreting the results of this study in the context of previously published, large series evaluating AJCC, it is important to note the different distribution of tumour sizes. In the present study, T3 and T4 tumours comprise 18% and 6% of the cases, compared with a large report by Kujala *et al*, in which T3 and T4 tumours made up 31% and 12% of the tumours included.⁴⁰

Table 4 Comparison of current study to prior reports evaluating the prognostic significance of LBD and GEP

	No of cases	Mean follow-up	Mean (median) tumour thickness	Mean (median) tumour LBD	No of metastatic events	Main conclusions
Current study	348	40.0 (37.0) months	4.9 (3.9) mm	10.0 (9.6) mm	41 (12%)	A simple four-group prognostic model combining GEP and LBD <12 mm vs ≥ 12 mm is more predictive than AJCC alone and similar to AJCC+GEP
Walter <i>et al</i>	339	30.8 (24.5) months	6.6 (6.1) mm	14.6 (14.6) mm	70 (20.6%)	LBD ≥ 12 mm at the time of treatment imparted a poorer prognosis for class 2 uveal melanomas
Correa <i>et al</i>	299	33 (32.2) months	5.6 (5.1) mm	11.9 (12.0) mm	25 (8%)*	GEP and LBD and independent prognostic factors for metastasis and metastatic death

*Reported metastatic death.
LBD, largest basal diameter.

If high-risk lesions are observed and significant growth occurs prior to treatment, this may allow the tumour more time to accumulate genetic alterations,³⁸ resulting in monosomy 3 or GEP class 2. On the other hand, the independent association of tumour size with metastatic rates and survival, while controlling for GEP^{23 24} and monosomy 3 status¹⁶ demonstrates that an increase in tumour size in and of itself following a period of growth results in a poorer prognosis irrespective of the currently available cytogenetic factors. Adequately controlling for lead-time bias, however, remains a challenge in this ever-evolving debate; we have chosen to report 3 and 5 years, rather than overall and median survival in an effort to minimise this bias.

Limitations of this study include the relatively small number of outcome events (41 patients developed metastasis (12%)). Based on GEP alone, approximately 54 patients included in this study would have been expected to metastasise over the study period. The lower-than-expected rate of metastasis in this cohort of patients is likely due to the importance of tumour size and the relatively small tumours included in our study compared with the studies validating GEP. Additionally, data regarding PRAME status was only available for 38% (132/348) of our patients given that it was only routinely performed after July 2017. As such, we were unable to draw any meaningful conclusions with regard to its impact on metastasis-free survival. In addition, the mean follow-up of 3 years limits extrapolation to longer follow-up when outcomes for class 1A and class 1B diverge. As at least some of the patients with short follow-up times will go on to develop metastatic disease, a long-term follow-up study is required to confirm the findings reported herein. Despite this, our mean follow-up (40 months) is longer than the two previously published, similarly sized studies (mean follow-up: 30.8²⁴ and 33²³ months) evaluating the same primary outcome (table 4). Large studies evaluating AJCC have reported metastasis in stage I and II patients at 5 years to be 5% and 17%, respectively.¹⁰ As the majority of our cohort was comprised of stage I (47%) and stage II (41%) lesions, our overall metastatic rate of 12% appears to be in keeping with previously published series. Our conclusions regarding the comparison of this four-group system to AJCC should be interpreted in the context of our study size and number of metastasis. Therefore, further studies are required to validate our findings on an independent dataset.

While addition of AJCC stage did not add independent prognostic value in this series, it is important to note that our study is much smaller than those validating AJCC,¹² and as such, may have been underpowered to find a significant result on multivariable analysis. Our four-group classification system does not include the extensive list of variables incorporated in LUMPO^{28 29} and PRiMeUM,³⁰ which aim to provide highly individualised prognostication. Rather, the benefit and purpose of our model lies in its inherent simplicity, making it an easy-to-use point-of-care tool to aide in uncomplicating patient-centred discussions about prognosis without a significant reduction in prognostic performance.

In summary, the findings of this study add to the current body of literature supporting the use of GEP as a well-validated prognostic test in UM with a high technical success rate, even in patients with small tumours. More importantly, our results support the additional predictive power of tumour size (LBD) and the potential for separation of patients into four prognostic groups for predicting metastasis-free survival, based on GEP class and LBD. Accurate prognostication plays an important role in the contemporary practice of personalised medicine, as it allows clinicians to provide tailored patient counselling, customised follow-up and metastatic surveillance, and stratification

within clinical trials. Further studies are required to validate the performance of this easy-to-use four-group classification system on an independent dataset.

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